The Overview of the Technology :

There is always a slight risk of damage to cells or tissue from being exposed to any radiation, including the low levels of radiation. X-rays are harmful to the tissues under long time exposure. C-MET has developed lead free X-ray shielding material as well as fabricated medical apron using these materials.

Salient features of the indigenously developed Lead free aprons compared to commercially available ones are:

- Light weight
- Lead Free (Non Toxic) and hence Eco-friendly
- Low cost

Specifications of the products:

Nanostructured hexagonal BBS having average diameter of around 100 nm. The X-ray absorption /attenuation provide the protection level greater than 0.25 mm thick lead (Pb) equivalence at 80 kVp.



Lead free X-ray absorbing apron developed at C-MET

The Level of Technology Readiness at C-MET

Technology Readiness level – 4 (TR-4) cum demonstration at Laboratory scale

The period of Training Under the Agreement

The period of training under the agreement will be of 2-3 months.

The Batch Level that will be Demonstrated if Technology is Offered to the Proposer etc

The technology will be demonstrated for 50 gm batch level of lead free, X-ray absorbing material and fabrication of one prototype apron.

The ToT package contains the following

1. Document containing raw material details
2. Processing details
3. Product specification
4. Test plan and procedure
5. Training